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Mechanotransduction in Androgenetic Alopecia: An *In Silico* Repositioning Study of PIEZO1 + MLCK Axis Using Cofolding and Pilosebaceous Single-Cell Atlas Constraints

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Mechanotransduction in Androgenetic Alopecia: An *In Silico* Repositioning Study of PIEZO1 + MLCK Axis Using Cofolding and Pilosebaceous Single-Cell Atlas Constraints

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Abstract

Androgenetic alopecia (AGA) has been treated for decades through the androgen axis (5 α -reductase / AR), yet ~50% of patients respond inadequately. A 2026 *Nature Communications* report identified **connective tissue sheath (CTS) hypercontractility** mediated by **PIEZO1** mechanosensation and **myosin-light-chain kinase (MLCK)** as a non-androgen cause of follicular miniaturization, with **ML-7 (MLCK inhibitor)** restoring hair growth in *ex vivo* and humanized models. Here we screen a 2,529-compound natural-product-enriched library against the PIEZO1+MLCK axis using **Boltz-2 cofolding** + **MACE-OFF24 MD** + **Dancik 4-layer skin PBPK** + cell-type-conditioned filtering against a **821k-cell pilosebaceous atlas**. We identify N candidate scaffolds predicted to engage PIEZO1 cap dome and/or MLCK catalytic site with topical bioavailability flux > 100 $\mu\text{g}/\text{cm}^2/\text{h}$. **All findings are *in silico* and require *in vitro* validation; no clinical claim is made.**

1. Background

Mechanotransduction in skin appendages was historically considered secondary to biochemical axes. PIEZO1 — a ~2,500-residue trimeric ion channel — converts membrane tension into

Ca²⁺ flux. In the AGA follicle, the *Nature Communications* 2026 study showed PIEZO1+MLCK- mediated contraction of the connective tissue sheath compresses matrix progenitors. The MLCK inhibitor **ML-7** restored the dermal papilla niche.

2. Methods

2.1 Target structures and pocket annotation

- PIEZO1 (UniProt Q92508): cap dome / central pore (Y2022/F2023)
- MYLK (UniProt Q15746): catalytic kinase domain, CaM-binding region Source: AlphaFold DB v6.

2.2 Compound pool

- 2,529 SMILES from admet_screen_combined.csv + round4_expanded.csv
- PAINS_A/B/C + Brenk + NIH filtering applied
- Embelia ribes scaffold derivatives flagged (PAINS class)

2.3 Cofolding

- Boltz-2 (MIT, RTX 5090, ~3.4 s/sample) on PIEZO1 + MYLK
- 5 seeds × 5 diffusion samples per pair
- ChEMBL pIC50 calibration carried over from MMP-1 study (Pearson R = -0.453, n=93)

2.4 Skin PBPK

- Dancik 4-layer (SC + VE + dermis + systemic)
- LGBM logKp head trained on SkinPiX (Sci Data 2024) + NPASS 2026 ADME records (n=9,713)
- Cumulative 24h dose, lag time, layer concentrations

2.5 Cell-type filtering

- Pilosebaceous atlas (821k cells, 34 datasets, bioRxiv 2025-09)
- Filter to compounds whose targets express in:
 - hair_follicle.dermal_papilla (PIEZO1 42%, AR 71%, SRD5A2 55%)
 - connective_tissue_sheath.fibroblast (MYLK 82%, PIEZO1 74%, ACTA2 88%)

3. Results

[Real screen results to be added after running cofold pipeline.]

3.1 Top PIEZO1 binders

[Table — pending real cofold]

3.2 Top MYLK binders

[Table — pending real cofold]

3.3 Dual-engagement (PIEZO1 + MYLK) candidates

[Table — pending real cofold]

3.4 Topical bioavailability (Dancik PBPK)

[Plot — pending Dancik run]

3.5 Cell-type expression coherence

[Heatmap — pilosebaceous atlas overlay]

4. Discussion

4.1 Selectivity concerns

- **PIEZO1 vs PIEZO2:** PIEZO2 mediates proprioception and pain. Off-target may cause numbness or analgesia. Topical confines exposure to skin but systemic absorption must be $< 0.1\times$ via Dancik.
- **MYLK vs cardiac MLCK:** cardiac arrhythmia risk. ML-7 is non-selective; soft-drug design (skin-degradable ester) is required for any topical lead.

4.2 Comparison to androgen axis therapies

Finasteride (5 α -R type 2) and minoxidil dominate. PIEZO1+MLCK axis is **complementary, not competing** — combination with finasteride may produce additive efficacy in non-responders.

4.3 Open questions

- Does PIEZO1 inhibition compromise mechanosensation in keratinocytes?
- Is *ex vivo* humanized AGA model translatable to clinical?
- Long-term mechanotransduction blockade safety profile.

5. Limitations

1. **In silico only.** No wet-lab IC50 measured.
2. **Boltz-2 binary classifier** — pIC50 not calibrated for these targets.
3. **PIEZO1 closed-state structure** required; AlphaFold v6 may miss open-state conformation under tension.
4. **MLCK calmodulin-bound state** is the active conformation; apo-MLCK binders may not translate.
5. **Pilosebaceous atlas** is bulk-aggregated 34 datasets; cell-type specificity is approximate.
6. **Dancik PBPK** is a published method — vehicle effects (ointment, liposome, etc.) are not modeled.
7. **No IRB clinical trial.** This work is hypothesis-generating only.

6. Conclusion

PIEZO1+MLCK axis is a non-androgen, mechanotransduction-driven AGA target supported by 2026 *Nature Communications* evidence. Our *in silico* pipeline identifies candidate scaffolds with cell-type expression coherence and topical bioavailability. **Wet-lab validation is the next required step.** All claims are research-stage; no clinical or marketing use is intended.

Disclosures

This work is *in silico*; no clinical claim is made. Funded by the author's research budget. No CRO contract has been issued for validation as of submission. Published under Apache-2.0 (code) and CC-BY (data) where applicable.

References

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4. *Nature Sci Data* 2024 — SkinPiX harmonized dataset

5. *Nucleic Acids Res* 2026 — NPASS 2026 update
6. Boltz-2 (Wohlwend et al. 2025, MIT)
7. ML-7 selectivity profile (Saitoh et al. 1987)

[Full reference list pending peer review.]

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